

Сетевые технологии

Лабораторная работа №5

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Информация

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Цель данной работы — построение простейших моделей сети на базе коммутатора и маршрутизаторов FRR и VyOS в GNS3, анализ трафика посредством Wireshark.



Рис. 1: Топология №1

```
VPCS>  
VPCS> ip 192.168.1.11/24 192.168.1.1  
Checking for duplicate address...  
PC1 : 192.168.1.11 255.255.255.0 gateway 192.168.1.1  
  
VPCS> save  
Saving startup configuration to startup.vpc  
. done  
  
VPCS> 
```

Рис. 2: Настройка ip на pc1-iemashkov

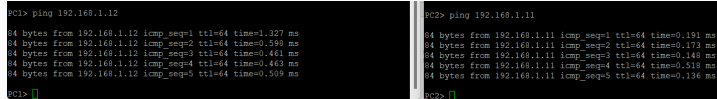
```
VPCS> ip 192.168.1.12/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.12 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> 
```

Рис. 3: Настройка ip на pc2-iemashkov

Выполнение лабораторной работы



The image shows two terminal windows side-by-side. The left window, labeled PC1, shows a ping command to 192.168.1.12 with five successful replies. The right window, labeled PC2, shows a ping command to 192.168.1.11 with five successful replies. Both windows show the standard ping output format: '84 bytes from [IP]: icmp_seq=[N] ttl=64 time=[X] ms'.

```
PC1> ping 192.168.1.12
84 bytes from 192.168.1.12 icmp_seq=1 ttl=64 time=1.327 ms
84 bytes from 192.168.1.12 icmp_seq=2 ttl=64 time=0.598 ms
84 bytes from 192.168.1.12 icmp_seq=3 ttl=64 time=0.461 ms
84 bytes from 192.168.1.12 icmp_seq=4 ttl=64 time=0.463 ms
84 bytes from 192.168.1.12 icmp_seq=5 ttl=64 time=0.509 ms
PC1> █
```

```
PC2> ping 192.168.1.11
84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=0.191 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=0.173 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=0.148 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=0.518 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=0.136 ms
PC2> █
```

Рис. 4: Взаимное пингование


```

1   9.056753 Private:66:b8:01 Broadcast ARP 64 Gratuitous ARP for 192.168.1.11 (Request)
2   9.056759 Private:66:b8:01 Broadcast ARP 64 Gratuitous ARP for 192.168.1.12 (Request)
3   9.158072 Private:66:b8:00 Broadcast ARP 64 Gratuitous ARP for 192.168.1.11 (Request)
4   9.145850 Private:66:b8:01 Broadcast ARP 64 Gratuitous ARP for 192.168.1.12 (Request)
5   7.2.054052 Private:66:b8:00 Broadcast ARP 64 Gratuitous ARP for 192.168.1.11 (Request)
6   7.2.056240 Private:66:b8:01 Broadcast ARP 64 Gratuitous ARP for 192.168.1.12 (Request)

```

```

# Frame 3: Packet; 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, Id 0
# Ethernet II, Src: Private:66:b8:00 (08:00:27:be:6d:80), Dest: Broadcast (::ffff::ff:ff:ff)
- Destination Broadcast (::ffff::ff:ff:ff)
    ....0.. ..... : LG bit: Locally administered address (this IS NOT the factory default)
    ....1... .. ... : LG bit: Group address (multicast/broadcast)
+ Source Private:66:b8:00 (08:00:27:be:6d:80)
    ....0.. ..... : LG bit: Globally unique address (factory default)
    ....0..... .. ... : LG bit: Individual address (unicast)
Type: ARP (Arphdr)
[Stream index: 2]
Padding: 0x00000000000000000000000000000000
Frame check sequence: ba0f080000 [unverified]
[Ethernet Status: Unsuccessful]

# Address Resolution Protocol (request/gratuitous ARP)
Hardware type: Ethernet (I)
Protocol type: IPv4 (0x0800)
Hardware size: 6
Protocol size: 4
Opcode: request (1)
Is gratuitous: true
Sender MAC address: Private:66:b8:00 (08:00:27:be:6d:80)
Send IP address: 192.168.1.11
Target MAC address: broadcast (::ffff::ff:ff:ff:ff)
Target IP address: 192.168.1.11

```

Рис. 5: ARP-запросы

Выполнение лабораторной работы

```
11 224.164189 192.168.1.12 192.168.1.11 ICMP 98 Echo (ping) request 16-0x8c9, seq=1/256, ttl=64 (reply in 12)
12 224.164493 192.168.1.11 192.168.1.12 ICMP 98 Echo (ping) reply 16-0x8c9, seq=1/256, ttl=64 (request in 11)
13 225.337375 192.168.1.12 192.168.1.11 ICMP 98 Echo (ping) request 16-0x8c9, seq=2/512, ttl=64 (reply in 14)
14 225.337884 192.168.1.11 192.168.1.12 ICMP 98 Echo (ping) reply 16-0x8c9, seq=2/512, ttl=64 (request in 13)
15 226.339488 192.168.1.12 192.168.1.11 ICMP 98 Echo (ping) request 16-0x8c9, seq=3/768, ttl=64 (reply in 16)
16 226.348179 192.168.1.11 192.168.1.12 ICMP 98 Echo (ping) reply 16-0x8c9, seq=3/768, ttl=64 (request in 15)
17 227.341181 192.168.1.12 192.168.1.11 ICMP 98 Echo (ping) request 16-0x8c9, seq=4/1024, ttl=64 (reply in 18)
18 227.341451 192.168.1.11 192.168.1.12 ICMP 98 Echo (ping) reply 16-0x8c9, seq=4/1024, ttl=64 (request in 17)
19 228.343728 192.168.1.12 192.168.1.11 ICMP 98 Echo (ping) request 16-0x8c9, seq=5/1280, ttl=64 (reply in 20)

*
* Frame 11: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface ..., id 8
* Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)
*   Destination Private_66:68:00 (00:50:79:66:68:00)
*     .... 0x00000000 .... = 16 bit: Globally unique address (factory default)
*     .... 0x00000000 .... = 16 bit: Individual address (unicast)
*   * Source: Private_66:68:01 (00:50:79:66:68:01)
*     .... 0x00000000 .... = 16 bit: Globally unique address (factory default)
*     .... 0x00000000 .... = 16 bit: Individual address (unicast)
*   Type: IPad (0x8000)
*   [Stream index: 4]
* Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11
*   0100 .... = Version: 4
*   .... 0101 = Header length: 20 bytes (5)
*   * Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
*   Total Length: 84
*   Identification: 0xc38c (51196)
*   * 0000 .... = Flags: 0x0
*   ... 0 0000 0000 0000 = Fragment Offset: 0
*   Time to Live: 64
*   Protocol: ICMP (1)
*   Header checksum: 0x2d85 [validation disabled]
*   [Header checksum status: Unverified]
*   Source Address: 192.168.1.12
*   Destination Address: 192.168.1.11
*   [Stream index: 0]
```

Рис. 6: ICMP-запросы

```
23 1884.695543 192.168.1.12 192.168.1.11 ECHO 98 Request
24 1884.695480 192.168.1.11 192.168.1.12 ECHO 98 Response
25 1885.697687 192.168.1.12 192.168.1.11 ECHO 98 Request
26 1885.697794 192.168.1.11 192.168.1.12 ECHO 98 Response
27 1886.708403 192.168.1.12 192.168.1.11 ECHO 98 Request
28 1886.708205 192.168.1.11 192.168.1.12 ECHO 98 Response

Frame 23: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface ..., id 0
Ethernet II, Src: Private_66:68:00 (08:00:29:66:68:00), Dst: Private_66:68:00 (08:00:29:66:68:00)
  Destination: Private_66:68:00 (08:00:29:66:68:00)
    .... 0: .... = 16 bit: Globally unique address (factory default)
    .... 0: .... = 16 bit: Individual address (unicast)
  Source: Private_66:68:00 (08:00:29:66:68:00)
    .... 0: .... = 16 bit: Globally unique address (factory default)
    .... 0: .... = 16 bit: Individual address (unicast)
  Type: IPv4 (0x0000)
  [Stream index: 4]
Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11
  0100 .... = Version: 4
  .... 0101 = Header length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total length: 64
  Identification: 0xcfb6 (53176)
  0000 .... = Flags: 0x0
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 64
  Protocol: UDP (17)
  Header checksum: 0x2779 (validation disabled)
  [Header checksum status: Unverified]
  Source Address: 192.168.1.12
  Destination Address: 192.168.1.11
  [Stream index: 0]
User Datagram Protocol, Src Port: 8176, Dst Port: 7
```

Рис. 7: UDP-запросы

Выполнение лабораторной работы

51 1976,924417	192.168.1.11	192.168.1.12	TCP	54.7 → 19958 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
52 1976,927023	192.168.1.12	192.168.1.11	TCP	66 19958 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSeq=1771898212 TSrc=0
53 1977,927199	192.168.1.12	192.168.1.11	TCP	76 [RST, Seq=0, Window=0, Len=0] Seq=0 Win=2920 Len=0 MSS=1460 TSeq=0
54 1977,928823	192.168.1.11	192.168.1.12	TCP	54.7 → 19958 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
55 1977,929512	192.168.1.12	192.168.1.11	TCP	66 19958 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSeq=1771898213 TSrc=0
56 1977,930412	192.168.1.12	192.168.1.11	ICMP	122 Request
57 1977,930501	192.168.1.11	192.168.1.12	TCP	54.7 → 19958 [ACK] Seq=1 Ack=57 Win=2920 Len=0
58 1977,931185	192.168.1.12	192.168.1.11	TCP	66 19958 → 7 [FIN, PSF, ACK] Seq=57 Ack=1 Win=2920 Len=0 TSeq=1771898213 TSrc=0
59 1977,931611	192.168.1.11	192.168.1.12	TCP	54.7 → 19958 [ACK] Seq=1 Ack=58 Win=2920 Len=0
+ Frame 54: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface ..., id 0				
+ Ethernet II, Src: Private_66:6d:80 (00:50:79:66:6d:80), Dst: Private_66:6d:81 (00:50:79:66:6d:81)				
+ Destination: Private_66:6d:81 (00:50:79:66:6d:81)				
+ 0b, = 16 bit: Globally unique address (factory default)				
+ 0b, = 16 bit: Individual address (unicast)				
+ Source: Private_66:6d:80 (00:50:79:66:6d:80)				
+ 0b, = 16 bit: Globally unique address (factory default)				
+ 0b, = 16 bit: Individual address (unicast)				
Type: IPv4 (0x0008)				
[Stream index: 4]				
+ Internet Protocol Version 4, Src: 192.168.1.11, Dst: 192.168.1.12				
+ 0100 = Version: 4				
+ 0101 = Header Length: 20 bytes (5)				
+ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)				
Total length: 40				
Identification: 0x0000 (53357)				
+ 0000 = Flags: 0x0				
+ ... 0 0000 0000 0000 = Fragment Offset: 0				
Time to Live: 64				
Protocol: TCP (6)				
Header Checksum: 0x26fb [validation disabled]				
[Header checksum status: Unverified]				
Source Address: 192.168.1.11				
Destination Address: 192.168.1.12				
[Stream index: 0]				
+ Transmission Control Protocol, Seq: 0, Dst Port: 19958, Seq: 0, Ack: 1, Len: 0				

Рис. 8: TCP-запросы

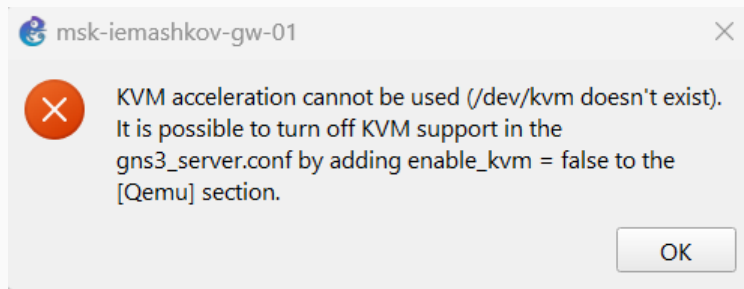


Рис. 9: Ошибка

Я получил навыки по построению простейших моделей сети на базе коммутатора и маршрутизаторов FRR и VyOS в GNS3, анализу трафика посредством Wireshark.